

The Scientific Genealogy of Dr. Ness B. Shroff

Nineteen generations of doctoral mentorship ascending to Nicolaus Copernicus, and the scholars guided in turn.

XIX
GENERATIONS1503
COPERNICUS, FERRARA1994
SHROFF, COLUMBIA51
DOCTORAL STUDENTS30
POSTDOCTORAL SCHOLARS1994–2026
YEARS SUPERVISING

I. THE LINE OF DESCENT

Each scholar was the doctoral advisor of the one beneath, read downward from Copernicus to Shroff in 1994. Where a scholar had two teachers the line divides and both routes are shown.

I FERRARA, 1503	Nicolaus Copernicus 1503 Doctorate in canon law, Ferrara · 1503 · studied Bologna, Padua, Kraków <i>"De revolutionibus orbium coelestium"</i> · displaced the Earth from the centre of the cosmos His own teachers — Leonhard von Dobschütz and Domenico Maria Novara — lie beyond this chart.			
II WITTENBERG, 1535	Georg Joachim Rheticus Magister artium, Universität Wittenberg · 1535 <i>The sole pupil of Copernicus; he persuaded his master to publish, and saw the book to press.</i> Second teacher: Johannes Volmar (Wittenberg 1515) — that line ends one step later.			
III LEIPZIG, 1567	Moritz Valentin Steinmetz Magister artium and Dr. med., Universität Leipzig · 1550, 1567 <i>"De Peste Capita Disputationis Ordinariae"</i> Second teacher: Johann Hoffmann — advisor unknown; the line ends there.			
IV LEIPZIG, 1582	Christoph Meurer Magister artium, Universität Leipzig · 1582 <i>"De Iride seu Arcu coelesti"</i> — on the rainbow			
V LEIPZIG, 1604	Philipp Müller Magister artium, Universität Leipzig · 1604 <i>Professor of mathematics at Leipzig.</i>			
VI LEIPZIG, 1650	Erhard Weigel Dr. phil., Universität Leipzig · 1650 <i>"De ascensionibus et descensionibus astronomicis dissertatio"</i> · taught the young Leibniz at Jena			
VII ALTDORF, 1666	Gottfried Wilhelm Leibniz Dr. jur., Universität Altdorf · 1666 <i>"Disputatio Inauguralis de Casibus Perplexis in Jure"</i> · co-inventor of the calculus			
	LEIBNIZ'S THREE OTHER TEACHERS <table><tr><td>Jakob Thomasius Magister, Universität Leipzig · 1643</td><td>Bartholomäus Schwendendörffer Universität Leipzig</td><td>Christiaan Huygens Académie royale des sciences, Paris · 1676</td></tr></table> <i>The Mathematics Genealogy Project records four teachers for Leibniz. Only Weigel's line, above, climbs on to Copernicus.</i>		Jakob Thomasius Magister, Universität Leipzig · 1643	Bartholomäus Schwendendörffer Universität Leipzig
Jakob Thomasius Magister, Universität Leipzig · 1643	Bartholomäus Schwendendörffer Universität Leipzig	Christiaan Huygens Académie royale des sciences, Paris · 1676		
VIII PARIS	Nicolas Malebranche Paris · philosopher and mathematician <i>The line divides here, and does not meet again until Poisson.</i>			

Malebranche taught two pupils; the descent runs through both, and the branches do not meet again until Poisson.

IX 1684 / 1687	THE BASEL LINE · VIA LAGRANGE		THE PARIS LINE · VIA LAPLACE	
	Jacob Bernoulli Universität Basel · 1684 <i>"Solutionem tertium problematis arithmetici, geometrici et astronomici"</i>		Pierre Varignon Collège Mazarin, Paris · 1687 <i>Statics and the mechanics of forces.</i>	
X 1694 / 1710	Johann Bernoulli Dr. med., Universität Basel · 1694 <i>"Dissertatio Inauguralis Physico-Anatomica de Motu Muscularum"</i>		Léonor Caron Collège Mazarin, Paris · 1710 <i>Taught from Varignon's own lecture notes.</i>	
XI 1726 / 1735	Leonhard Euler Universität Basel · 1726 <i>"Dissertatio physica de sono"</i>		Jean Le Rond d'Alembert Collège Mazarin, Paris · 1735 <i>The wave equation; co-editor of the Encyclopédie.</i>	
XII TURIN / PARIS	Joseph-Louis Lagrange no formal degree · Turin and Berlin <i>Taught by Euler through their correspondence.</i>		Pierre-Simon Laplace Paris · 1769 <i>Celestial mechanics; analytic probability.</i>	
— the two lines reunite —				
XIII PARIS, 1800	Siméon Denis Poisson École Polytechnique, Paris · 1800 <i>Student of both Lagrange and Laplace — the two lines reunite here.</i>			
COLLATERAL LINE · NOT ANCESTORS	Jean-Baptiste Joseph Fourier Ph.D.		Gustav P. Lejeune Dirichlet Hon., Universität Bonn · 1827	
COLLATERAL LINE · NOT ANCESTORS	Rudolf O. S. Lipschitz Dr. phil., Universität Berlin · 1853			
Fourier, Dirichlet and Lipschitz were Poisson's students — relatives of the line, not steps within it.				

II. DOCTORAL STUDENTS · THE FIFTY-ONE

Advised to the Ph.D. at Purdue and The Ohio State University, in order of completion, 1998–2026.

Junyi Li PH.D. FEB. 1998 <i>"Intercellular Coordination Schemes for Improving Spectrum Utilization in Wireless Cellular Networks"</i> Co-founder, Flarion Technologies; now VP of Research, Qualcomm Inc.	Jinwoo Choe PH.D. AUG. 1998 <i>"Queueing Analysis of High-Speed Networks with Gaussian Traffic Models"</i> Associate Professor, Sogang University, Seoul [IN MEMORIAM · 2009]	Paul Salama PH.D. AUG. 1999 <i>"Error Concealment in Encoded Images and Video"</i> Professor of EE, IUPUI, Indianapolis, Indiana	Ting-Li Ling PH.D. DEC. 1999 <i>"Traffic Management and Control in ATM Networks"</i> Senior Engineer, Onepage.com, San Jose, California
Suresh Kalyanasundaram PH.D. AUG. 2000 <i>"Call Level and Class Level Quality of Service in Multiservice Networks"</i> Bell Labs Fellow, Nokia, Bengaluru, India	Mingbo Xiao PH.D. APR. 2002 <i>"Resource Management in Power-Controlled Cellular Wireless Networks"</i> Professor of EE, Xiamen University, China	Xin Liu PH.D. AUG. 2002 <i>"Opportunistic Scheduling in Wireless Communication Networks"</i> Professor of Computer Science, University of California, Davis	Han S. Kim PH.D. MAR. 2003 <i>"Queueing Analysis of Loss Ratio and End-to-End Delay Distribution"</i> Samsung Master (Executive), Samsung Electronics, Seoul
Do Young Eun PH.D. MAY 2003 <i>"Analysis, QoS Estimation, and Decomposition of Large Networks"</i> Professor of ECE, North Carolina State University	Dongyu Qiu PH.D. SEP. 2003 <i>"Distributed Flow Control for Next Generation Networks"</i> Professor of EE, Concordia University, Montreal	Jang Won Lee PH.D. JUL. 2004 <i>"Non-convex Optimization and Resource Allocation in Communication Networks"</i> Professor of EE, Yonsei University, Seoul	Xiaojun Lin PH.D. AUG. 2005 <i>"Simplification of Network Dynamics in Large Systems"</i> Professor, School of ECE, Purdue University
Longbi Lin PH.D. NOV. 2006 <i>"Energy Efficient Routing of Multi-Hop Networks"</i> Managing Director, Deutsche Bank, Singapore	Gaurav Sharma PH.D. NOV. 2006 <i>"Design and Analysis of Wireless and Wireless-Wireline Networks"</i> Vice President, D. E. Shaw & Co., New York	Sungoh Kwon PH.D. APR. 2007 <i>"Routing for Multihop Wireless Networks"</i> Professor, University of Ulsan, South Korea	Issa Khalil PH.D. JUL. 2007 <i>"Mitigation of Control and Data Attacks in Ad hoc and Sensor Networks"</i> Research Director, Qatar Computing Research Institute
Yan Wu PH.D. MAY 2007 <i>"Energy Management in Sensor Networks"</i> Member of Technical Staff, Microsoft Inc.	Roman Chertov PH.D. MAY 2008 <i>"A Device Independent Router Model"</i> Director of Software Engineering, Arista Networks, Santa Clara	Gagan R. Gupta PH.D. OCT. 2009 <i>"Delay Efficient Control Policies for Wireless Networks"</i> Senior Principal Member of Technical Staff, San Ramon, California	Sarah Sellke PH.D. JAN. 2010 <i>"Analytical Characterization of Internet Security Attacks"</i> Associate Teaching Professor of Computer Science, Northeastern University
Joohwan Kim PH.D. MAR. 2010 <i>"Improving Performance of Wireless Networks Using Anycast"</i> Principal Engineer, Samsung Mobile, Seoul	Abdallah Khreishah PH.D. MAR. 2010 <i>"Analysis and Design of Interession Network Coding in Communication Networks"</i> Professor of ECE, New Jersey Institute of Technology	Sugumar Murugesan PH.D. AUG. 2010 <i>"Opportunistic Scheduling Using Channel Memory in Markov-modeled Wireless Networks"</i> Data Scientist and Technical Lead, Johnson Controls	Srikanth Hariharan PH.D. DEC. 2011 <i>"Communication-Efficient Convergecasting for Data Fusion in Wireless Sensor Networks"</i> Senior Manager, R&D, VMware, Palo Alto
Bo Ji PH.D. DEC. 2012 <i>"Design of Efficient Resource Allocation Algorithms for Wireless Networks"</i> Professor of Computer Science, Virginia Tech	Shuang Li PH.D. MAY 2013 <i>"Managing Dynamic Spectrum Access Under Uncertainty in Cognitive Radio Networks"</i> Head of Product, ML Platform, Liffott	Shengbo Chen PH.D. AUG. 2013 <i>"Resource Allocation and Control in Communications and Cyberphysical Networks with Renewable Energy"</i> Professor, School of Software, Nanchang University, China	Zhoujia Mao PH.D. AUG. 2013 <i>"Approximation Algorithms Design for Resource Management in Communication Networks"</i> Performance and Capacity Engineer, Facebook, Menlo Park
Wenzhuo Ouyang PH.D. APR. 2014 <i>"Scheduling in Wireless Networks with Limited and Imperfect Information"</i> Manager, Amazon, Seattle	Swapna Buccapatnam PH.D. NOV. 2014 <i>"Control of Large Scale Networked Systems Under Uncertainty"</i> Principal Inventive Scientist, AT&T Labs, Middletown, New Jersey	Yousi Zheng PH.D. APR. 2015 <i>"Scheduling and Design in Cloud Computing Systems"</i> Senior Software Engineer, LinkedIn, San Francisco	Yang Yang PH.D. APR. 2015 <i>"Throughput Characterization and Algorithmic Design in Wireless Networks"</i> Staff Software Engineer, Google, San Diego
Irem Eryilmaz (Koprulu) PH.D. MAY 2016 <i>"Spread of Information and Influence over Dynamic Networks"</i> Associate Professor of Practice, The Ohio State University	Ming Zhang PH.D. AUG. 2016 <i>"Game Theory and Algorithm Design in Network Security and Smart Grid"</i> Senior Staff Engineer and Manager, Google, Mountain View	Fei Wu PH.D. FEB. 2019 <i>"Ultra-Low Delay in Complex Computing in Networked Systems"</i> Senior Inventive Scientist, AT&T Labs Research, New Jersey	Sinong Wang PH.D. FEB. 2019 <i>"Coded Computation for Speeding up Distributed Machine Learning"</i> Principal Research Scientist, Facebook AI, Seattle
Zhenzhi Qian PH.D. APR. 2019 <i>"Designing High-Performance Scheduling Policies in Multi-Channel Multi-antenna Networks"</i> Software Engineer, Google, Seattle	Jiashang Liu PH.D. APR. 2020 <i>"Resource Allocation and Energy Management in Green Network Systems"</i> Software Engineer, BigQuery ML, Google, Kirkland	Fang Liu PH.D. APR. 2020 <i>"Efficient Online Learning with Bandit Feedback"</i> Research Scientist, Facebook, Menlo Park	Sherif ElAzzouni PH.D. APR. 2020 <i>"Algorithmic Design for Low Latency Communication in Wireless Networks"</i> Senior Engineer, Qualcomm, San Diego
Xingyu Zhou PH.D. DEC. 2020 <i>"Load Balancing in Heavy Traffic: Theory and Algorithms"</i> Associate Professor of ECE, Wayne State University, Detroit	Ahmed Bedewy PH.D. APR. 2021 <i>"Optimizing Data Freshness in Information Update Systems"</i> Senior Engineer, Qualcomm, Bridgewater, New Jersey	Wenbo Ren PH.D. APR. 2021 <i>"Active Learning for Ranking from Noisy Observations"</i> Research Scientist, Facebook, Menlo Park	Guocong Quan PH.D. DEC. 2021 <i>"Designing Caching Mechanisms for Distributed and Dynamic Caching Systems"</i> Assistant Professor of Computer Science and Engineering, Sun Yat-sen University, Guangzhou
Guidan Yao PH.D. JUN. 2022 <i>"Information Freshness and Delay Optimization in Unreliable Wireless Systems"</i> Algorithm and Simulation Engineer, Nokia, Chicago	Yuntian Deng PH.D. JUN. 2022 <i>"Operation of Multi-modal Transportation Networks: Optimization and Reinforcement Learning"</i> Applied Scientist, Amazon, Seattle	Jayanth Regatti PH.D. DEC. 2022 <i>"Learning at the Edge under Resource Constraints"</i> Data and Applied Scientist, Microsoft Ads, Bellevue	Jiayu Pan PH.D. JUL. 2023 <i>"Optimizing Age of Information and Beyond in Wireless Networks"</i> Assistant Professor of Software Engineering, Hangzhou, China
Sunjung Kang PH.D. AUG. 2024 <i>"Enhancing Remote Tracking Systems in IoT Networks: Balancing Timely Updates and Resource Constraints"</i> Assistant Professor, Gwangju Institute of Science and Technology, South Korea	Hao Chen PH.D. JUL. 2025 <i>"Detection, Adaptation, and Optimization of Time-Varying Stochastic Systems in Wireless Networks"</i> Research Scientist, Meta, Menlo Park	Yining Li PH.D. JUL. 2026 <i>"Probably Efficient Algorithms for Scalable Reinforcement Learning"</i> Research Scientist, Meta, Menlo Park	

III. POSTDOCTORAL SCHOLARS & RESEARCH ASSOCIATES

Thirty mentored since 1996. Of the twenty-eight who have moved on, twenty-five held faculty appointments.

Zhenghu Gong [VISITING] 1986–87 National University of Defense Technology, China	Dong Sik Kim [VISITING] 1986–89 Professor, Hankuk University, South Korea	Wonjong Noh [VISITING] 2005–07 Principal Research Engineer, Samsung Advanced Institute of Technology	Changhee Joo 2005–10 Professor of Computer Science, Korea University
Jin-Ghoo Choi 2008–10 Associate Professor, Yeungnam University, South Korea	Jian Tian 2009–10 Senior Staff Engineer, Alibaba DAMO Academy	Yoora Kim 2010–13 Assistant Professor of Mathematics, University of Ulsan	Zizhang Zhang 2010–14 Associate Professor of Computer Science, Tulane University
Jia (Kevin) Liu 2010–14 Associate Professor of ECE, The Ohio State University	MHR Khouzani 2011–13 Senior Lecturer, Victoria University of Wellington	Yin Sun 2011–14 Godbold Endowed Associate Professor of ECE, Auburn University	Joohyun Lee 2010–14 Associate Professor of EE, Sogang University
Xiaowen Gong 2015–16 Godbold Associate Professor of ECE, Auburn University	Morteza Hashemi 2015–19 Associate Professor of EECS, University of Kansas	Jaewook Kwak 2016–18 Software Engineer, Broadcom	Rahul Singh 2016–18 Assistant Professor of ECE, Indian Institute of Science
Gourav Saha 2020–21 Assistant Professor, Mahindra University, Hyderabad	Doyup Kim 2021–22 Kyungnam University, South Korea	Arnob Ghosh 2021–23 Assistant Professor of ECE, New Jersey Institute of Technology	Peizhong Ju 2021–26 Assistant Professor of CS, University of Kentucky
Ming Shi 2022–24 Assistant Professor of CS, SUNY Buffalo	Sen Lin 2022–24 Assistant Professor of CS, University of Houston	Ziyue Luo 2022–26 Associate Professor, Wuhan University, China	Mohamed A. Abd-Elmagid 2023–25 Research Assistant Professor of ECE, Virginia Tech
Chengzhang Li 2023–26 Associate Professor of CS, Shanghai Jiao Tong University	Yuchen Liang 2023–26 Assistant Professor, Luddy School, Indiana University Indianapolis	Kumar Saurav 2024–26 Assistant Professor of Electrical Engineering, Shiv Nadar Institution of Eminence	Jihyeon Yun 2024–26 Assistant Professor of ECE, Kangwon National University, South Korea
Tasmeen Zamaan Ornee CURRENT Postdoctoral scholar	Hongbo Li CURRENT Postdoctoral scholar		

Reading this chart. A scientific genealogy traces doctoral advisors rather than blood relation. Poisson studied under both Lagrange and Laplace, so the descent from Malebranche runs by two equal routes that reunite in 1800. The chart is capped at Copernicus because above him the record thins into mentorship reconstructed by scholars rather than degrees conferred. **Sources:** the Mathematics Genealogy Project, the original Scientific Genealogy tree, and Prof. Shroff's curriculum vitae.